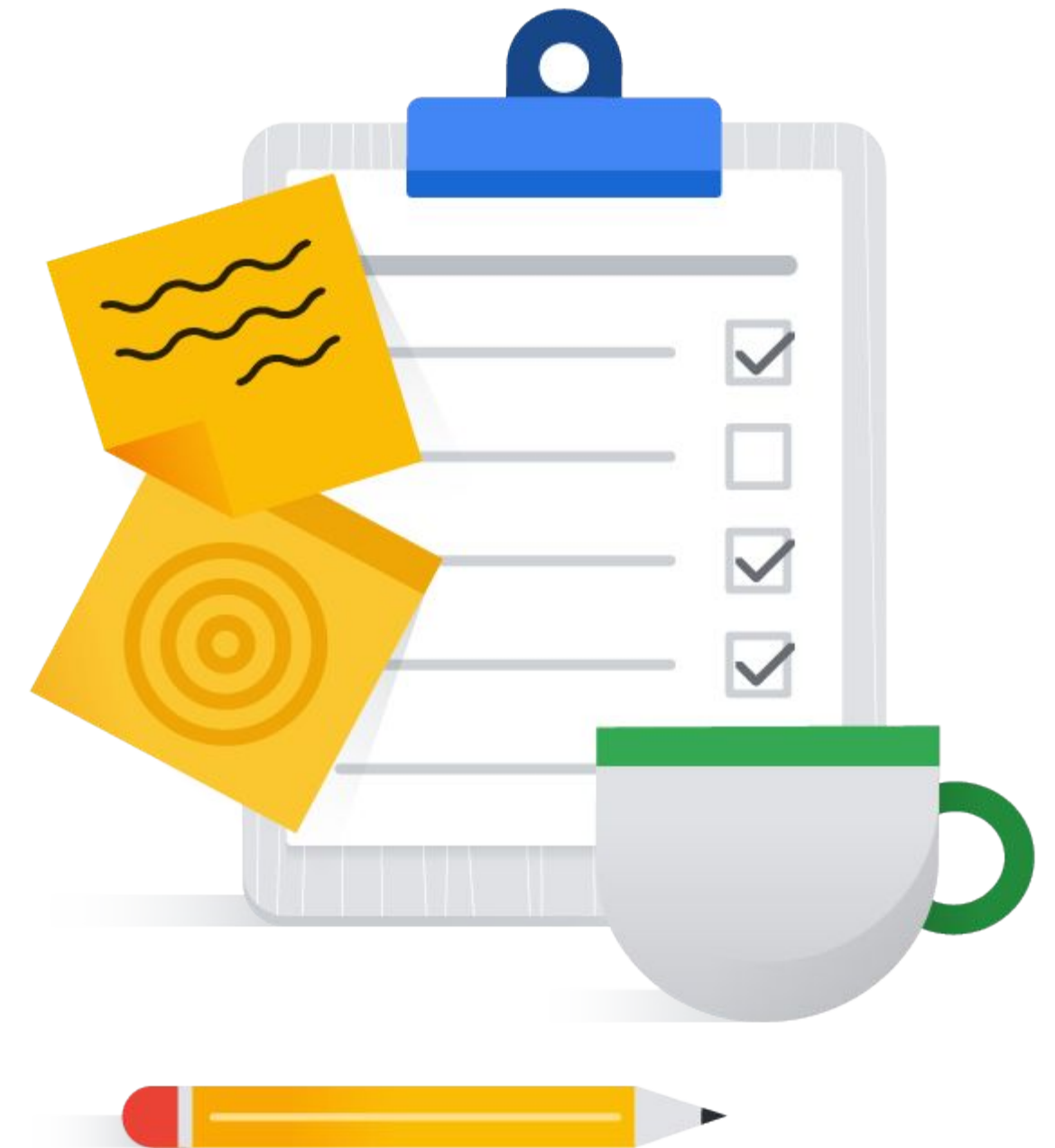




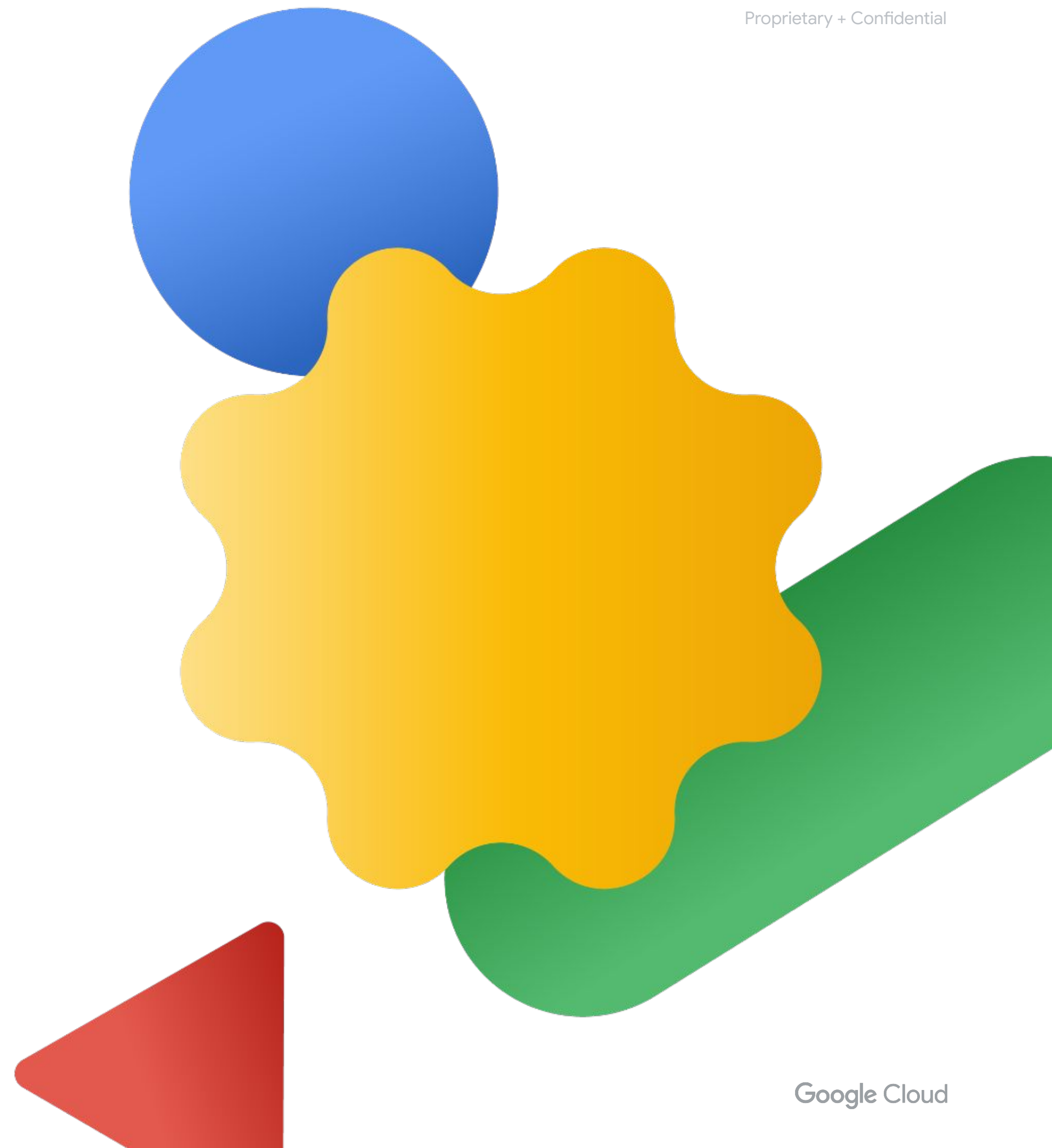
Introducing Agent Development Kit (ADK)

Module objectives

01	Identify how AI agents significantly impact operations.
02	Compare approaches to building AI agents on Google Cloud.
03	Explain the functionality of ADK.
04	Explore the features and benefits of ADK.



AI agents



The rise of AI Agents

33%

Enterprise software will incorporate AI agents

15%

AI agents will automate daily work decisions

Customer agents

Employee agents

Creative agents

Data agents

Code agents

Security agents

AI Agents are interactive partners that **help answer complex questions and complete tasks.**

AI Agents



AI Agents

Data science	Employee productivity	Research	Developer
Get insights faster by automating data-related tasks			

AI Agents

Data science

Get insights faster by
automating
data-related tasks

Employee productivity

Help train and
onboard
Complete daily tasks

Research

Developer

AI Agents

Data science

Get insights faster by
automating
data-related tasks

Employee productivity

Help train and
onboard
Complete daily tasks

Research

Quickly answer
questions using
proprietary data

Developer

AI Agents

Data science

Get insights faster by
automating
data-related tasks

Employee productivity

Help train and
onboard
Complete daily tasks

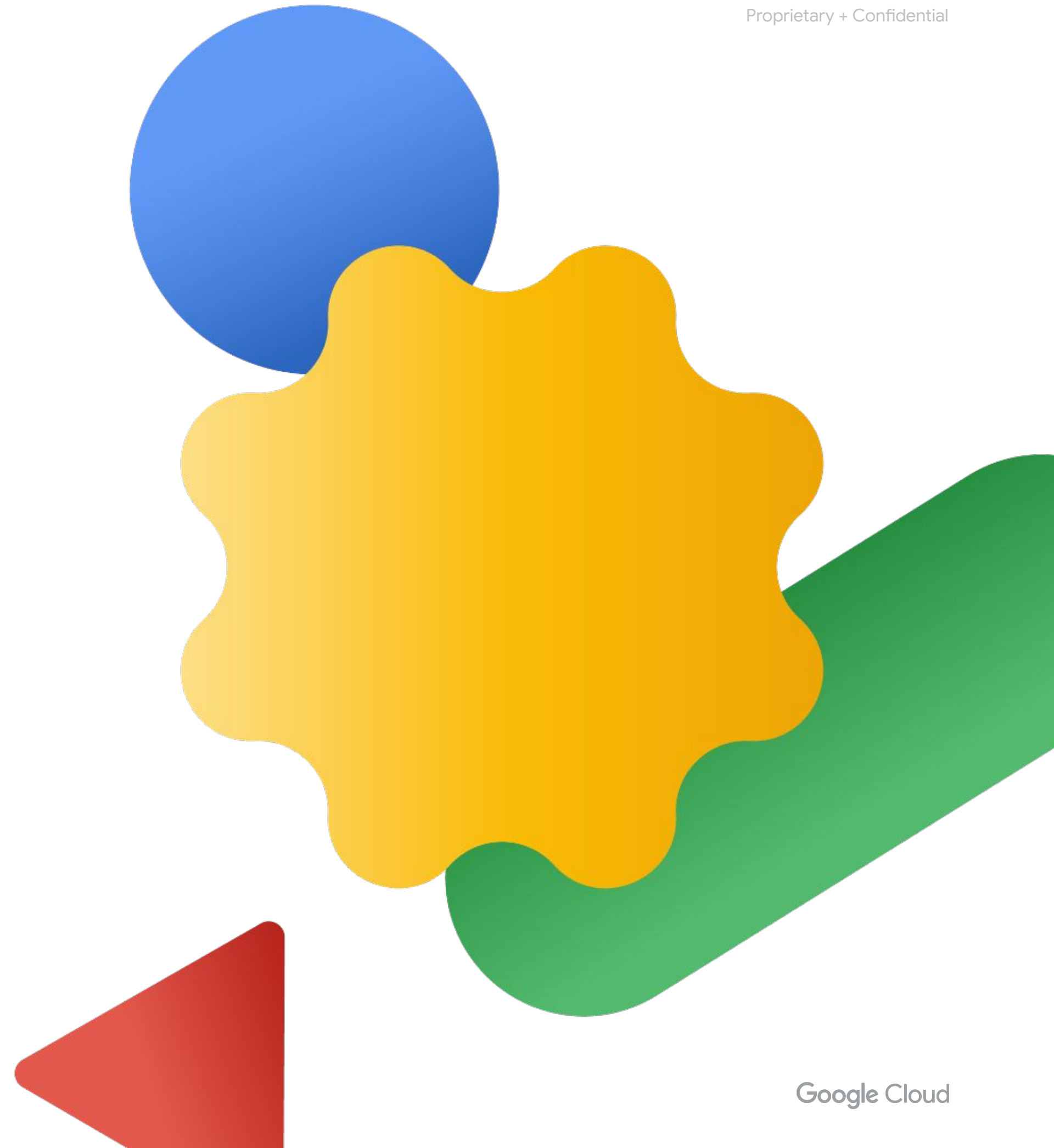
Research

Quickly answer
questions using
proprietary data

Developer

Support completing
software
development tasks

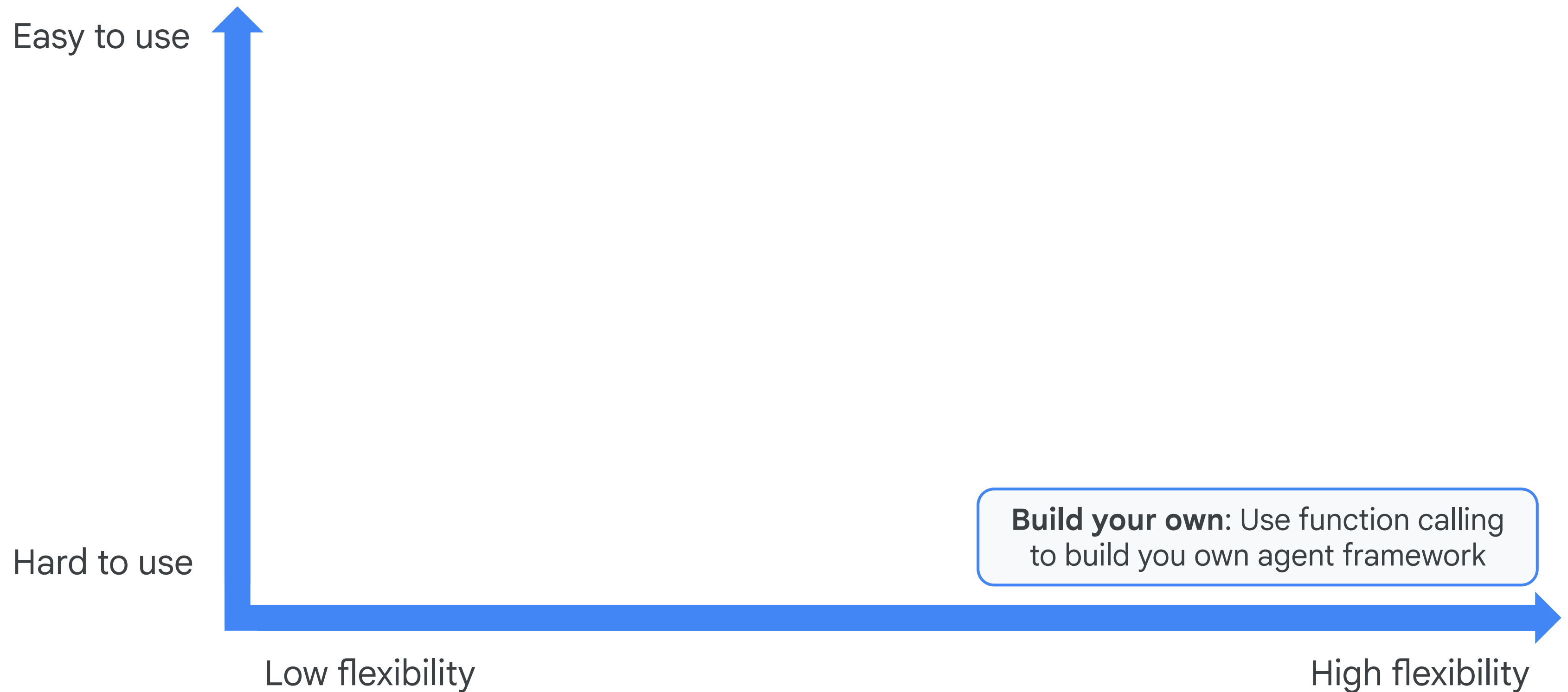
Building agents



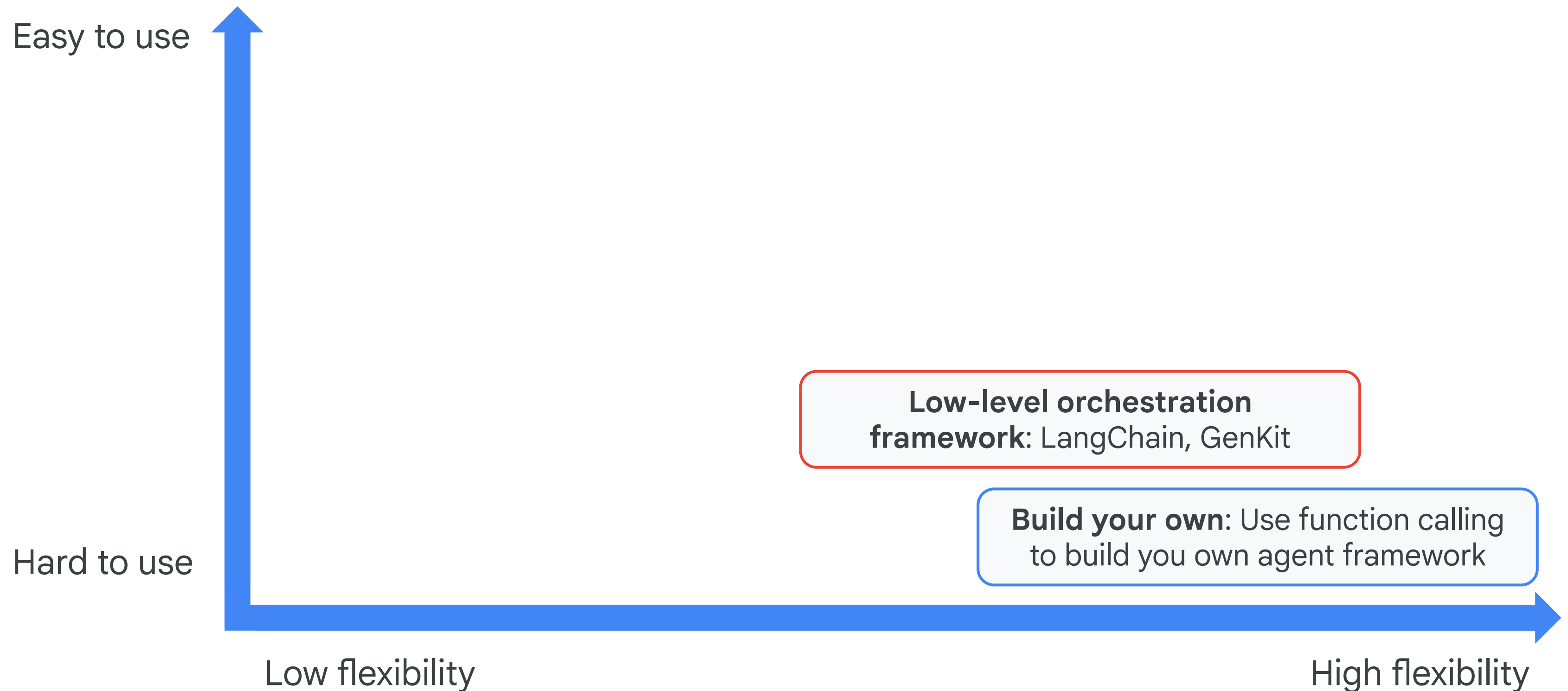
Higher abstraction makes systems easier to build



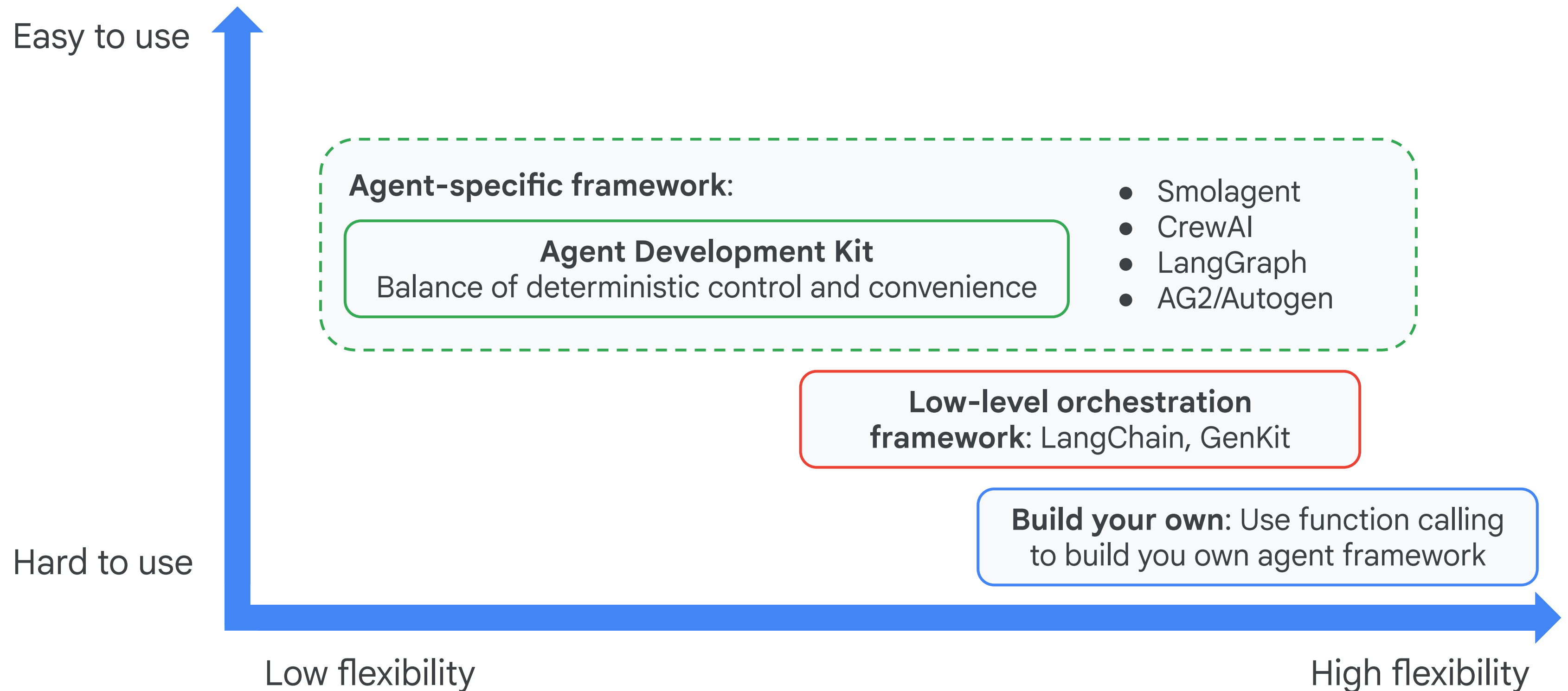
Higher abstraction makes systems easier to build



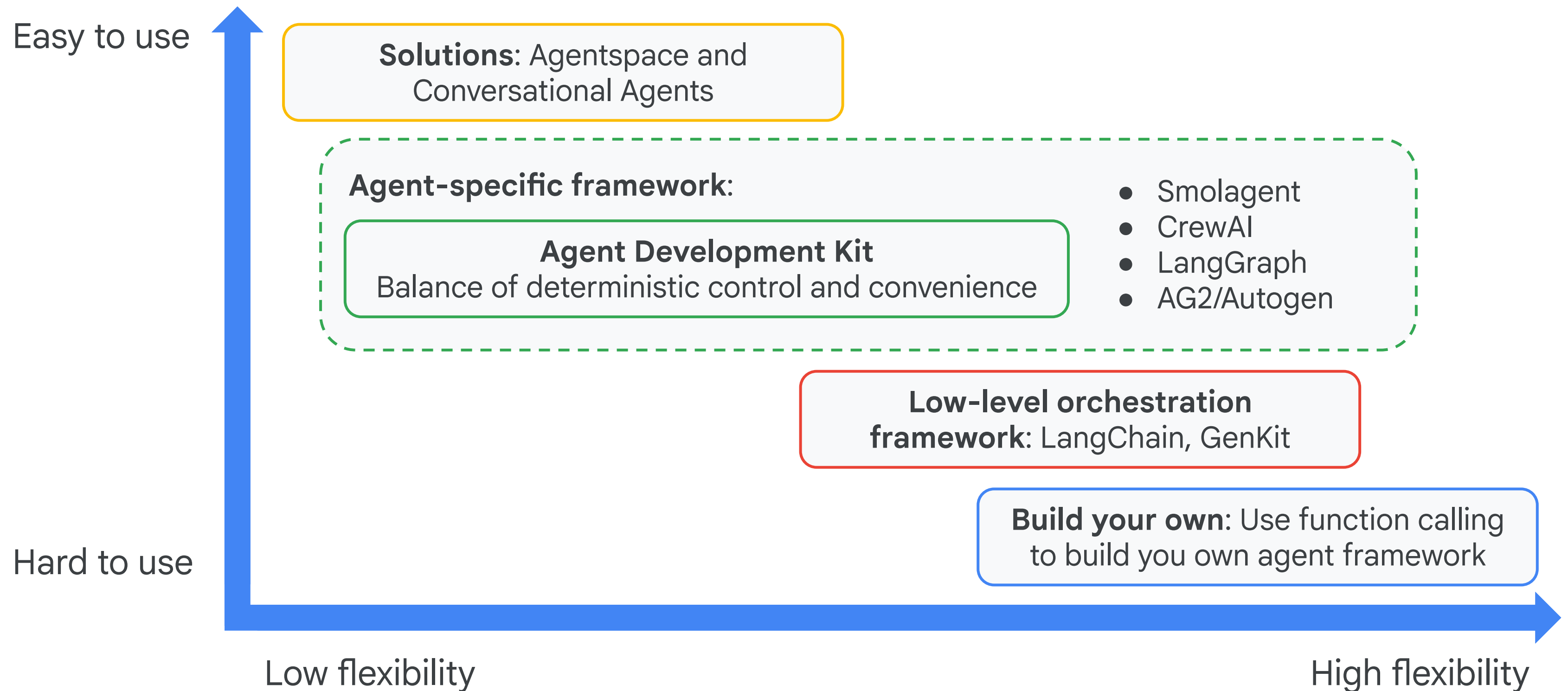
Higher abstraction makes systems easier to build



Higher abstraction makes systems easier to build



Higher abstraction makes systems easier to build



Building an agent

- ✓ Conversational agents
- ✓ Agentspace
- ✓ Custom solutions with Agent Engine
- ✓ Use Agent Development Kit with Agent Engine

Building an agent

- ✓ Conversational agents
- ✓ Agentspace
- ✓ Custom solutions with Agent Engine
- ✓ Agent Development Kit with Agent Engine

External-facing conversational agents that can integrate with human support teams and existing telephony and communication platforms

Building an agent

- ✓ Conversational agents
- ✓ Agentspace
- ✓ Custom solutions with Agent Engine
- ✓ Agent Development Kit with Agent Engine

Internal search to accelerate knowledge exchange throughout your organization, across your drives, chat, mail, ticketing platforms, databases, including AI assistant support

Building an agent

- ✓ Conversational agents
- ✓ Agentspace
- ✓ Custom solutions with Agent Engine
- ✓ Agent Development Kit with Agent Engine

Build agents from scratch (custom)

- Google Gen AI SDK or LangChain

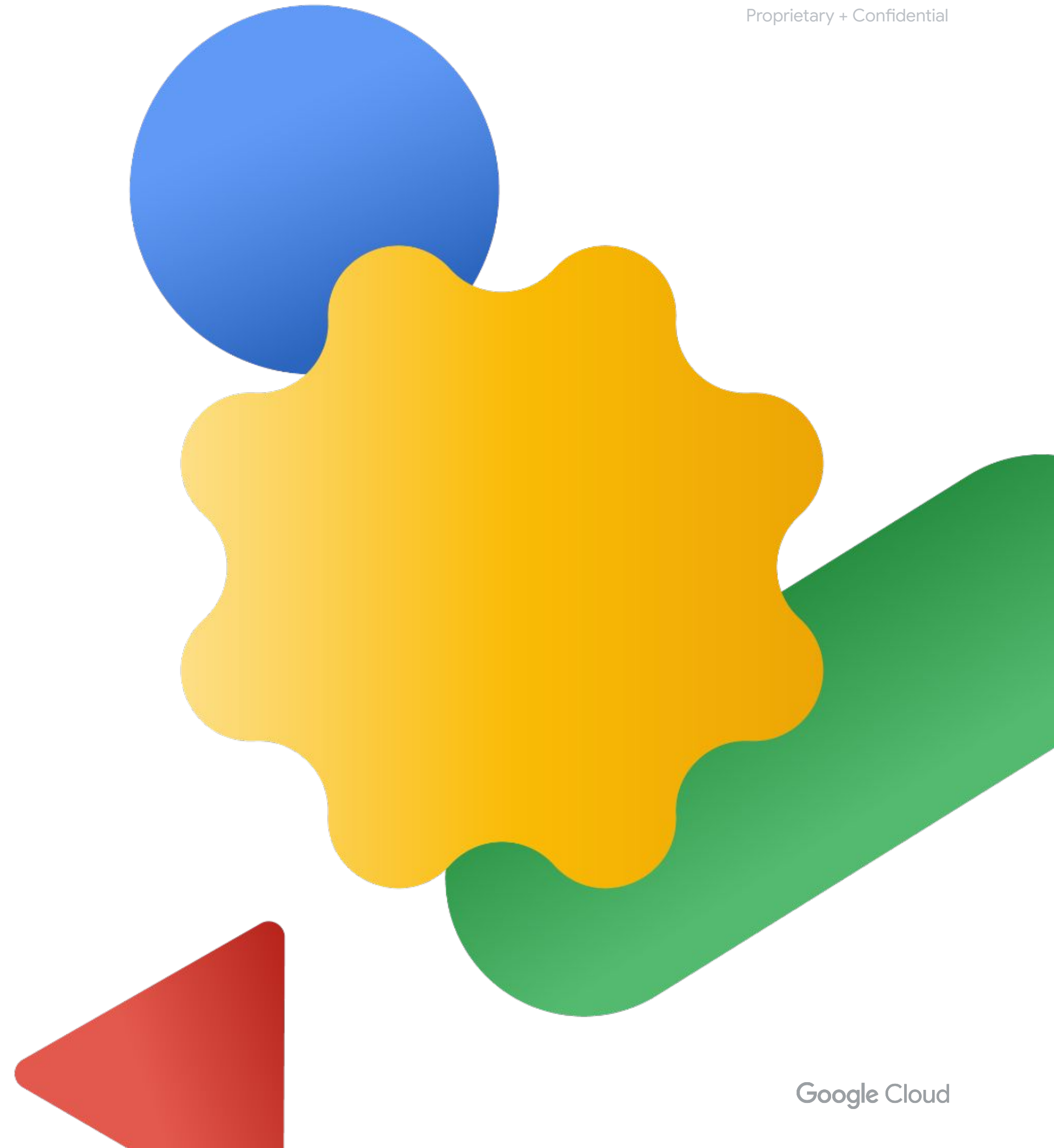
Building an agent

- ✓ Conversational agents
- ✓ Agentspace
- ✓ Open source solutions or from scratch with Agent Engine
- ✓ Agent Development Kit with Agent Engine

Custom development with support for communication between agents through conversation history and shared state

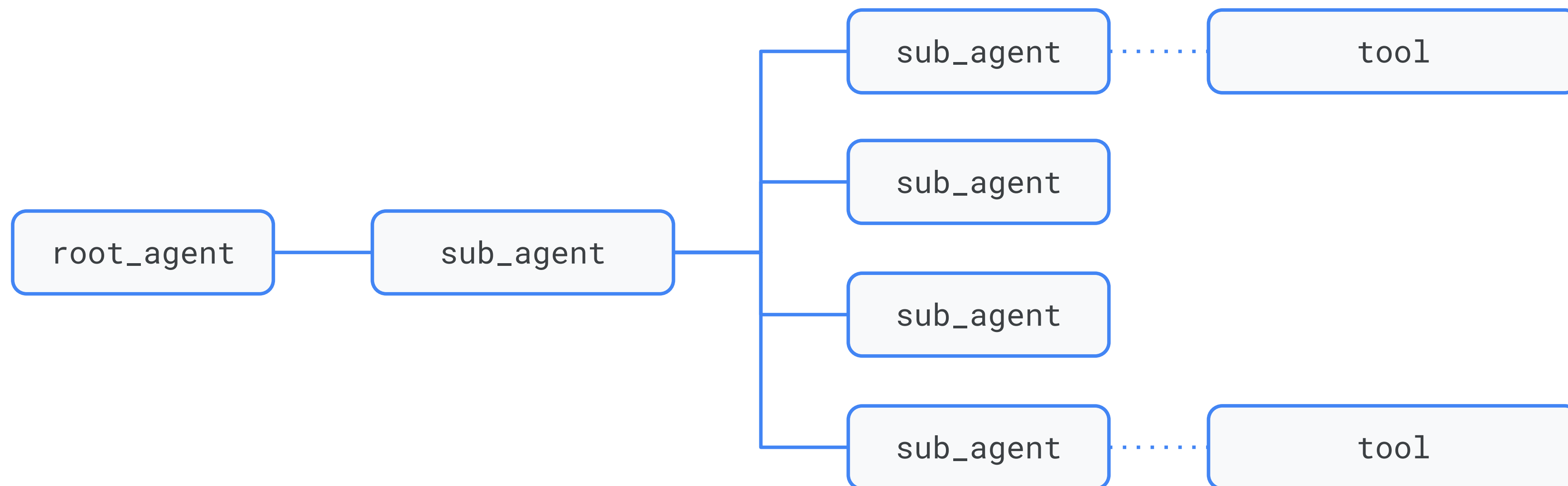
- A set of developer-friendly tools to create agents within applications
- Code-centric approach
- No AI expertise required

Agent Development Kit (ADK)



A **client-side, Python SDK** enabling developers to quickly **build and customize multi-agent systems.**

Multi-agent systems



What does the ADK include?

Multi-agent systems

Prebuilt, custom, and
community tools

Support for evaluation

A convenient local
development UI with
debugging tools

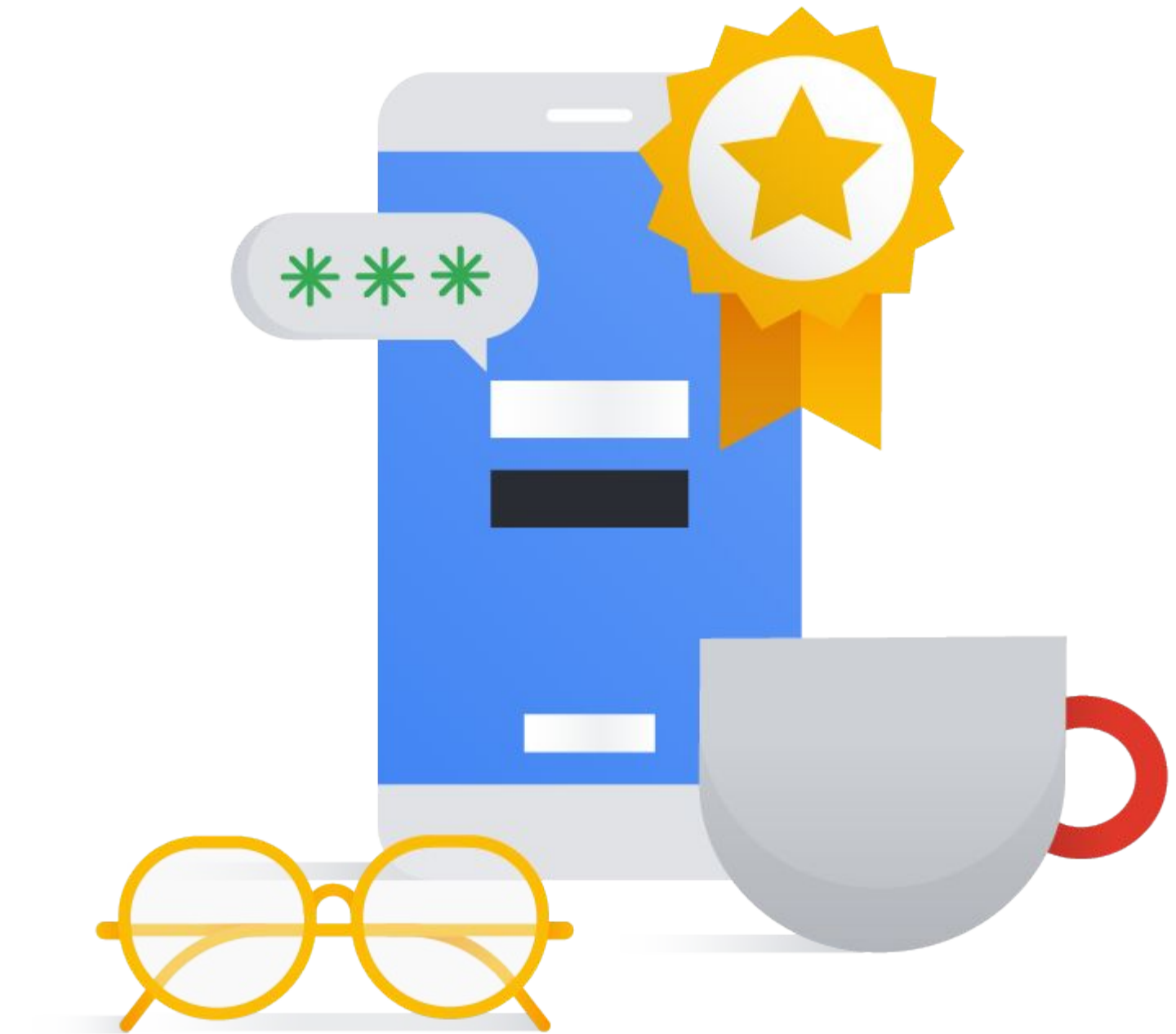
Callbacks to invoke
functions throughout
the flow

Session memory for
stateful conversations

Collaboration on
documents with
shared artifacts

Can deploy to
Agent Engine for
fully-managed
agent infrastructure

Now let's do a short
quiz to **check your
knowledge!**



Quiz | Question 01

Question

What is Agent Development Kit primarily designed for?

- A. Business users with no coding experience
- B. AI/ML architects training new model architectures
- C. Developers building multi-agent systems
- D. End users interacting with conversational agents

Quiz | Question 01

Answer

What is Agent Development Kit primarily designed for?

- A. Business users with no coding experience
- B. AI/ML architects training new model architectures
- C. Developers building multi-agent systems
- D. End users interacting with conversational agents



Quiz | Question 02

Question

Regarding tools, what does Agent Development Kit primarily offer?

- A. Prebuilt, custom, and community tools
- B. Only prebuilt tools from Google
- C. Only custom tools built from scratch
- D. No tools; it focuses only on agent orchestration

Quiz | Question 02

Answer

Regarding tools, what does Agent Development Kit primarily offer?

- A. Prebuilt, custom, and community tools
- B. Only prebuilt tools from Google
- C. Only custom tools built from scratch
- D. No tools; it focuses only on agent orchestration



Quiz | Question 03

Answer

What is a key benefit of Agent Development Kit regarding infrastructure decisions?

- A. It requires developers to manage their own infrastructure
- B. It only supports on-premise infrastructure
- C. You can avoid infrastructure decisions by deploying to Agent Engine.
- D. It requires extensive knowledge of cloud infrastructure

Quiz | Question 03

Answer

What is a key benefit of Agent Development Kit regarding infrastructure decisions?

- A. It requires developers to manage their own infrastructure
- B. It only supports on-premise infrastructure
- C. It eliminates the need for infrastructure decisions by deploying to Agent Engine
- D. It requires extensive knowledge of cloud infrastructure



Google Cloud